TASK TELANGANA · CLOUD SOFTWARE DEVELOPER TRAINING

# Module 03: AWS Core Services & Infrastructure Master Handbook

### The 3-Day Technical Deep Dive: Compute, Storage Triad, Databases, VPC Networking, Containers (Docker/ECR/ECS) & IaC CloudFormation

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**TRAINER OVERVIEW:** This master guide is compiled specifically for the Nalgonda training cohort. It covers the full technical breadth required for enterprise cloud engineering and AWS Certified Developer (DVA-C02) standards. Concepts are grounded in **The Grand Vehicle Metaphor** and accompanied by an explicit **Service Decision Architecture**.

## 1. The Grand Vehicle Metaphor: The Cloud as an Automobile Fleet

To eliminate confusion among dozens of cloud services, think of AWS as an on-demand vehicle rental and logistics agency:

| AWS Service | Vehicle Metaphor | Real-World Function |
|---|---|---|
| **Amazon EC2** | **Car Engine & Chassis** | Rent an engine on-demand. Choose horsepower (Instance Type: *t3.micro*), model (AMI: *Amazon Linux*), and key (Key Pair *.pem*). |
| **Amazon EBS** | **Car's Personal Trunk** | Local hard drive bolted to 1 car chassis in a parking garage (AZ). Data stays safe when engine turns off. |
| **Amazon EFS** | **Shared Fleet Roof Cargo** | Shared roof rack accessed simultaneously by 100+ cars across multiple garages via NFS protocol. |
| **Amazon S3** | **Global Container Depot** | Infinite worldwide warehouse outside the vehicle. Store objects in buckets; reachable via web URLs (HTTPS). 11 9's durability. |
| **Amazon VPC** | **Gated Industrial Campus** | Private gated corporate park with secure perimeter walls. Only authorized vehicles enter. |
| **Internet Gateway** | **Main Highway Gate** | Two-way highway opening connecting public showroom lanes to the internet. |
| **NAT Gateway** | **One-Way Security Escort** | Allows private vault cars to fetch fuel/software without letting outside internet cars in. |
| **Security Group** | **Car's Personal Bodyguard** | **Stateful:** If a passenger is approved to enter, they can step back out automatically. |
| **Network ACL** | **Highway Border Toll Booth** | **Stateless:** Checks license plates on entry AND explicitly checks license plates on exit. |

## 2. Service Requirements & Decision Matrix: What is Required & When to Use Which Service

To make architectural decisions with confidence, memorize both the technical prerequisites and the exact decision triggers:

| Service | What is Required (Prerequisites) | When to Choose It (Decision Trigger) |
|---|---|---|

| | | |
|---|---|---|
| **Amazon EC2** | • AMI OS template<br>• Instance Type (vCPU/RAM)<br>• VPC Subnet & Key Pair (.pem)<br>• Security Group with open ports | Need full administrative root access over the OS, custom software daemons, or monolithic architectures. |
| **Amazon EBS** | • Must be in **exact same AZ** as EC2<br>• Formatted in Linux (ext4) and mounted | OS boot drives and high-performance transactional database storage requiring <1ms low latency. |
| **Amazon EFS** | • VPC Mount Targets in each AZ<br>• Security Group allowing Port 2049 (NFS) | Multiple EC2 servers (100+) need concurrent shared read/write access to a single filesystem (CMS, web uploads). |
| **Amazon S3** | • Globally unique bucket name (3-63 chars)<br>• Block Public Access ON<br>• IAM / Bucket Policy permissions | Static website hosting, user images/videos, backups, analytics lakes. 99.999999999% durability. |
| **Amazon RDS** | • DB Subnet Group across 2+ AZs<br>• Security Group allowing DB Port (5432/3306) | Relational SQL databases requiring ACID transactions, complex JOINs, and automated Multi-AZ failover. |
| **Amazon Aurora** | • DB Subnet Group<br>• Cluster endpoint configuration | Enterprise high-throughput SQL; 5x faster than MySQL, 6-way storage replication, Serverless v2. |
| **DynamoDB** | • Table name & Partition Key (PK)<br>• Capacity Mode (On-Demand / Provisioned) | Serverless NoSQL Key-Value store requiring single-digit millisecond latency at any scale (shopping carts, IoT). |
| **ElastiCache (Redis)** | • Cache Subnet Group<br>• Security Group Port 6379 from app servers | Sub-millisecond in-memory caching to eliminate SQL database query bottlenecks and store user session states. |
| **Internet Gateway** | • Attached to VPC<br>• Route table points 0.0.0.0/0 to IGW | Public-facing web servers or Application Load Balancers needing direct 2-way internet traffic. |
| **NAT Gateway** | • Placed in a **Public Subnet**<br>• Allocated an **Elastic IP (EIP)** | Private subnet instances needing outbound-only internet (for OS patches/pip install) while remaining hidden from incoming hackers. |
| **Amazon ECS (Fargate)** | • Docker Image in ECR<br>• Task Definition (CPU/RAM)<br>• IAM Execution Role & VPC Subnets | Running continuous Docker containers (web APIs, worker daemons) without managing underlying EC2 virtual machines. |
| **AWS Lambda** | • Function code / Container image (max 10GB)<br>• IAM Execution Role<br>• Event Trigger (S3, API GW) | Event-driven tasks executing in <15 minutes. Scales to zero automatically with $0 idle cost. |
| **CloudFormation** | • Declarative YAML/JSON template<br>• Valid Resources section | Automating infrastructure deployment from Git. Catching manual out-of-band edits via **Drift Detection**. |

## 3. Infrastructure Drift Detection & Remediation

**WHAT IS DRIFT?** Drift occurs when an engineer makes manual out-of-band changes directly in the AWS Console (e.g. resizing an EC2 instance or opening Port 22 in a Security Group) without updating the CloudFormation template.

**HOW TO CATCH & REMEDIATE DRIFT:**

1. Run **Detect stack drift** in the CloudFormation Console.
2. Review the drift report: AWS flags drifted properties in red side-by-side with template expected values.
3. **Remediation:** Either revert the manual console change back to match the template, OR update the CloudFormation YAML template to adopt the new configuration and deploy via a Change Set.

## 4. Student Zero-Dollar Sunset Checklist

1. Stop practice EC2 instances (pauses compute charges).
2. Delete orphan EBS volumes in 'available' status.
3. Delete custom CloudFormation stacks to tear down child resources cleanly.
4. Release any allocated Elastic IPs.
5. Check AWS Billing Dashboard weekly to ensure $0.00 balance.